

The Expanse Psyche

Specification and Deployment Guide

SubThought Corporation
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1 Overview

The Expanse is a Psyche: the interface between a GIL mind and the three-dimensional physical reality that biological organisms currently inhabit. It mediates two flows — percepts traveling inward from the physical world to the mind's mechanisms, and actuations traveling outward from the mind's mechanisms to the physical world. The Expanse unifies the action surfaces of multiple robot platforms — Figure, Optimus, NAO, Unitree G2, and others — into a single coherent interface.

The Expanse encompasses the planet Earth, the oceans, inner space, and outer space. It is the three-dimensional world that robot variants will navigate, perceive, and act upon. Microphones, cameras, depth sensors, LIDAR, haptic sensors, force/torque transducers, IMUs, and GPS receivers provide multimodal percepts from the Expanse to the mind of the Learner.

All percepts from the Expanse carry :Modality Expanse. The :Channel slot selects the sensory domain. The :Data slot names the parsable structure type. The :Content slot carries the idiom.

2 The Percept Relation

The Percept relation from totality.theory is canonical. It does not change for the Expanse.

```
(relation Percept
  :M                ; lexified monad
  :Modality         ; Expanse for this device
  :Channel          ; sensory sub-channel
  :Address nil      ; sensor URI
  :Data nil         ; format type
  :Content nil      ; idiom {:Slot value ...}
)
```

2.1 Slot Semantics

Slot	Meaning	Expanse Examples
:Modality	Which Psyche delivered this percept.	Expanse (always)
:Channel	Sensory sub-channel.	Visual, Auditory, Haptic, Proprioceptive, Spatial, Olfactory, Communication, System
:Address	Sensor address or device URI.	"ros2://figure-01/camera/head", "ros2://nao-03/mic/left"
:Data	Data format type.	camera-frame, depth-map, transcription, contact, joint-state, odometry, gps-fix
:Content	Idiom with structured data.	{:Width 640 :Height 480 :Format rgb8 :FrameId "head_camera"}
:Moment	When the percept was received.	\@m{2026060100000000}

3 Percepts by Channel

All percepts from the Expanse have :Modality Expanse. The :Channel identifies the sensory domain.

3.1 Channel Visual

:Data	:Address	:Content (idiom)	Platforms
camera-frame	ros2://.../camera/head	{:Width 640 :Height 480 :Format rgb8 :FrameId ...}	All
depth-map	ros2://.../depth/front	{:Width 640 :Height 480 :Format 32FC1 :MinRange 0.2}	Figure, Optimus, G2
point-cloud	ros2://.../lidar/front	{:Width 1024 :PointStep 16 :Fields {...}}	Figure, Optimus, G2
object-detected	nil	{:Class "cup" :Confidence 0.92 :BBox {...} :Distance 1.2}	All
face-detected	nil	{:FaceId ... :Confidence 0.88 :Expression neutral}	All

3.2 Channel Auditory

:Data	:Address	:Content (idiom)	Platforms
audio-sample	ros2://.../mic/array	{:SampleRate 16000 :Channels 4 :Format pcm_s16le}	All
transcription	nil	{:Text "hello" :IsFinal yes :Confidence 0.95 :Speaker ...}	All
speech-detected	nil	{:Direction 45.0 :Energy 0.72 :FrameId "mic_array"}	NAO, Figure, G2

3.3 Channel Haptic

:Data	:Address	:Content (idiom)	Platforms
contact	ros2://.../hand/right	{:Regions {:Palm yes :Fingers ...} :Pressure 2.4}	Figure, Optimus, G2
force-reading	ros2://.../wrist/right/ft	{:Force {:X 0.5 :Y -0.2 :Z 9.8} :Torque {...}}	Figure, Optimus
collision	nil	{:LinkName ... :Force 15.2 :Unit "N"}	All

3.4 Channel Proprioceptive

:Data	:Address	:Content (idiom)	Platforms
joint-state	ros2://.../joint_states	{:Joints [{:Name ... :Position 0.5 :Velocity 0.0} ...]}	All
imu-reading	ros2://.../imu	{:Orientation {:Roll ... :Pitch ... :Yaw ...} :LinearAcceleration {...}}	All
battery-state	ros2://.../battery	{:Voltage 12.4 :Percentage 0.73 :Charging no}	All
balance-state	nil	{:CenterOfMass {:X 0.01 :Y -0.02 :Z 0.85} :Stable yes}	All

3.5 Channel Spatial

:Data	:Address	:Content (idiom)	Platforms
odometry	ros2://.../odom	{:Position {:X 3.2 :Y 1.5} :LinearVelocity {:X 0.5}}	All
gps-fix	ros2://.../gps	{:Latitude 37.77 :Longitude -122.42 :Altitude 10.2}	Outdoor
pose	ros2://.../pose	{:Position {:X ... :Y ... :Z ...} :Orientation {:X ... :W ...}}	All
occupancy-grid	nil	{:Width 200 :Height 200 :Resolution 0.05 :FrameId "map"}	All (SLAM)

3.6–3.8 Olfactory, Communication, System

Olfactory: gas-reading, air-quality (future platforms). Communication: speech-heard, gesture-seen, touch-signal, display-tap. System: connection-status, safety-event, mode-change, diagnostic. See the Expanse.md companion file for complete idiom definitions.

4 Common Content Idiom Slots

Slot	Type	Count	Notes
:FrameId	string	25+	Coordinate frame of reference. Universal.
:Unit	literal	20+	Measurement unit (rad, N, m, C, deg).
:Width / :Height / :Duration	integer	12	Dimensional measure.
:Confidence	real	10	Certainty of detection.
:Position / :Orientation	idiom	8	Pose in 3D space.
:Force / :Torque / :Pressure	idiom/real	5	Physical force measurements.
:PersonId / :Speaker	string	5	Identity of a detected person.

5 Actuations

The Expanse defines 35 actuation literals across 8 domains.

Domain	Count	Literals
Locomotion	8	walk, turn, stop, step, stand, sit, crouch, navigate
Gaze	2	look, track
Manipulation	7	reach, grasp, release, push, pull, lift, place
Gesture	4	point, wave, nod, shake-head
Communication	4	speak, emote, display, play-sound
Motor Control	4	joint-move, pose-move, trajectory, set-stiffness, set-led
Capture	3	capture-photo, start-recording, stop-recording
System	2	e-stop, set-mode

6 Cross-Cutting Structure

6.1 The Four Tuple Types

```

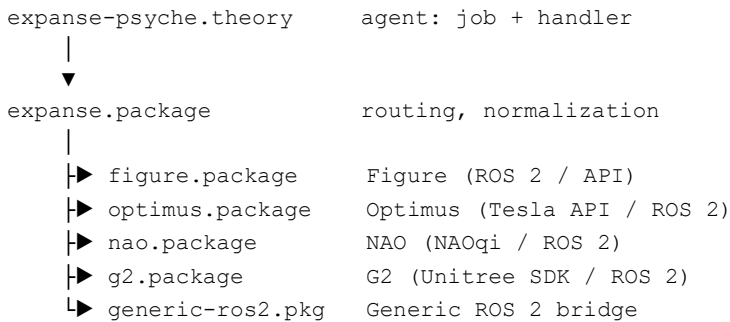
PERCEPT:  [PERCEPT :Modality Expanse :Channel C :Address A :Data D :Content {...}
             :Moment M :Token T]
ATTEMPT:    [ATTEMPT :Action A :Parameters {...} :Token T :By M]
RESULT:     [RESULT  :Action A :Status S :Reason R :Moment M :Token T]
URGE:       [URGE    :Need N :Source S :Delta D :Moment M :Token T]

```

6.2 Actuation → Percept Pairings

:Action	Produces :Data
walk, turn, step, navigate	odometry, joint-state, balance-state
look, track	camera-frame, object-detected, face-detected
reach, grasp, release	joint-state, contact, force-reading
push, pull, lift, place	force-reading, joint-state, contact
speak	audio-sample (echo), transcription
joint-move, pose-move, trajectory	joint-state
capture-photo	camera-frame
navigate	odometry, pose, occupancy-grid
e-stop	safety-event

7 Architecture



8 Deployment File Set

File	Category	Description
figure.package	Adapter	Figure humanoid protocol
optimus.package	Adapter	Tesla Optimus protocol
nao.package	Adapter	NAO NAOqi/ROS 2 protocol
g2.package	Adapter	Unitree G2 protocol
generic-ros2.package	Adapter	Generic ROS 2 bridge
expanse.package	Unified Layer	Routing, normalization, convenience
expanse-psyche.theory	Theory	Single agent: job + handler
Expanse.daicho	Configuration	Platforms, routing, polling, urges
expanse.suites	Tests	Full test coverage

9 Platform Asymmetries

Figure:

41+ DOF, dexterous hands, force/torque at wrists, whole-body dynamic locomotion.

Optimus:

GPS for outdoor navigation, actuator-level telemetry, Tesla vision pipeline.

NAO:

4-mic array with sound localization, capacitive head sensors, chest tablet, NAOqi behavior library.

G2:

Agile legged locomotion, LIDAR-based SLAM, Unitree high-speed motor control.

All share:

Cameras, IMU, joint feedback, walking, speaking, grasping (varying DOF), ROS 2 compatibility.

10 Etymology

Expanse (Latin expansum, from expandere, “to spread out, to unfold”) — first attested in English in 1667, coined by John Milton in *Paradise Lost* as his translation of the Hebrew *raqia* from Genesis. The word captures the physical world’s most defining quality: it is vast, open, three-dimensional, and stretches out in every direction without visible boundary.

Where the Aether emphasizes the medium through which signals propagate (the internet), the Expanse emphasizes the territory through which bodies move (the physical world). Together they cover the full range of worlds a GIL mind can inhabit.

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Auditory — see §3.2

Communication — see §3.7

Expanse — see §1, §10

Figure — see §8, §9

G2 — see §8, §9

Grasp — see §5

Haptic — see §3.3

Joint-state — see §3.4

Locomotion — see §5

Manipulation — see §5

NAO — see §8, §9

Navigate — see §5

Olfactory — see §3.6

Optimus — see §8, §9

PERCEPT — see §2, §6

Proprioceptive — see §3.4

ROS 2 — see §8, §9

Spatial — see §3.5

System — see §3.8

Visual — see §3.1

